業　績　調　書

フリガナ 氏　　名	周阳

１．修士論文または主な研究論文の概要

２．業績リスト（主な研究論文、主な学会発表、著書等を年代順に記載すること）

１．修士論文または主な研究論文の概要（Abstract）

Hematologic malignancies are critically threatening to human health, and timely detection of hematological malignancies is crucial to the treatment of the disease. Bone marrow cell morphological examination is a key step in the diagnosis of hematological malignancies and an important basis for the prevention and screening of hematological malignancies. It provides an important reference standard for the treatment and recovery of patients. At present, manual detection of bone marrow microscopic images is time-consuming and labor-intensive, and is susceptible to the subjective influence of pathologists during the detection process, causing deviations in the detection results. Deep learning technology can realize efficient automated detection of bone marrow microscopic images and solve the problems existing in manual detection. Deep learning model training relies on a large number of high-quality labeled samples and bone marrow microscopic images will lead to staining differences due to factors such as different laboratory equipment. These two problems greatly affect the performance of deep learning methods. Therefore, this article takes bone marrow microscopic images as the main research object, and the research work includes the following aspects.

This thesis collects and constructs a larger scale bone marrow micrographic image dataset, which comes from 256 confirmed patients, including bone marrow micrographic images of four types of hematologic malignancies, it is a bone marrow micrographic image dataset with larger data size and more types of diseases at this stage.

Aiming at the problem of staining differences in bone marrow micro-images, this thesis proposes a semi-supervised staining normalization method based on the self-attention mechanism with dual decoders, which stains and normalizes the bone marrow micro-images to be used in the subsequent classification task. The method designs a dual-decoder generator structure that enables the model to efficiently learn unlabeled source domain images in an unsupervised manner by enforcing the consistency of the images generated by the two decoders. The model also incorporates a self-attention mechanism and an auxiliary feature extraction network to improve the model coloring performance and robustness. In this thesis, the effectiveness of this thesis's method is verified in two publicly available medical image datasets and bone marrow microscopy image dataset by two types of evaluation metrics, namely, image quality and color consistency. The effectiveness of the method as an image preprocessing step is evaluated in a bone marrow microscopy dataset classification task.

Aiming at the problems of manual detection and difficult access to image data, this thesis proposes a semi-supervised pseudo-label-enhanced generative adversarial network-based hematologic malignancies classification method. Unlike previous methods that rely on cell detection and counting, this thesis's method is based on the bone marrow microscopic images as a whole and achieves the semi-supervised classification task for four common hematologic malignancies. The method in this thesis combines generative adversarial networks and pseudo-label generation techniques to better learn the semantic features of the images through SimCLR in the pre-training phase. This thesis proposes the MixTeacher pseudo-label generation method which introduces a teacher-student network on top of MixMatch to improve the accuracy and robustness of the classification model. This thesis also designs the classifier structure of a global classifier combined with a local classifier and introduces the spatial self-attention mechanism and pixel self-attention mechanism to extract richer features of bone marrow microscopy images. Through experimental validation, this thesis's method outperforms other semi-supervised classification methods in terms of classification accuracy with limited labeling and achieves the classification of four types of hematologic malignancies.

※は記入しない。

２．業績リスト（主な研究論文、主な学会発表、著書等を年代順に記載すること）

2024 Zhou Yang, Shao Hong, Cui WenchengSemi-Supervised Stain Normalization Network Based on Self-Attention Mechanism. [J] Journal of Biomedical Engineering(under review)

2024 Yang Zhou*, Hong Shao. PLE-GAN: Pseudo-label enhanced generative adversarial networks for medical image classification[C]. 2024 3rd International Symposium on Computer Applications and Information Systems(ISCAIS 2024). Xiamen:SPIE,2023, 224031123531928922.

2024 Invention Patent. A semi supervised staining normalization method. FYJC24FI03705.

2023 Computer software copyright. Bone marrow aspiration smear staining normalization software. 2023SR0815338

※は記入しない。

Abstract

Hematologic malignancies are critically threatening to human health, and timely detection of hematological malignancies is crucial to the treatment of the disease. Bone marrow cell morphological examination is a key step in the diagnosis of hematological malignancies and an important basis for the prevention and screening of hematological malignancies. It provides an important reference standard for the treatment and recovery of patients. At present, manual detection of bone marrow microscopic images is time-consuming and labor-intensive, and is susceptible to the subjective influence of pathologists during the detection process, causing deviations in the detection results. Deep learning technology can realize efficient automated detection of bone marrow microscopic images and solve the problems existing in manual detection. Deep learning model training relies on a large number of high-quality labeled samples and bone marrow microscopic images will lead to staining differences due to factors such as different laboratory equipment. These two problems greatly affect the performance of deep learning methods. Therefore, this article takes bone marrow microscopic images as the main research object, and the research work includes the following aspects.

This thesis collects and constructs a larger scale bone marrow micrographic image dataset, which comes from 256 confirmed patients, including bone marrow micrographic images of four types of hematologic malignancies, it is a bone marrow micrographic image dataset with larger data size and more types of diseases at this stage.

Aiming at the problem of staining differences in bone marrow micro-images, this thesis proposes a semi-supervised staining normalization method based on the self-attention mechanism with dual decoders, which stains and normalizes the bone marrow micro-images to be used in the subsequent classification task. The method designs a dual-decoder generator structure that enables the model to efficiently learn unlabeled source domain images in an unsupervised manner by enforcing the consistency of the images generated by the two decoders. The model also incorporates a self-attention mechanism and an auxiliary feature extraction network to improve the model coloring performance and robustness. In this thesis, the effectiveness of this thesis's method is verified in two publicly available medical image datasets and bone marrow microscopy image dataset by two types of evaluation metrics, namely, image quality and color consistency. The effectiveness of the method as an image preprocessing step is evaluated in a bone marrow

microscopy dataset classification task.

Aiming at the problems of manual detection and difficult access to image data, this thesis proposes a semi-supervised pseudo-label-enhanced generative adversarial network-based hematologic malignancies classification method. Unlike previous methods that rely on cell detection and counting, this thesis's method is based on the bone marrow microscopic images as a whole and achieves the semi-supervised classification task for four common hematologic malignancies. The method in this thesis combines generative adversarial networks and pseudo-label generation techniques to better learn the semantic features of the images through SimCLR in the pre-training phase. This thesis proposes the MixTeacher pseudo-label generation method which introduces a teacher-student network on top of MixMatch to improve the accuracy and robustness of the classification model. This thesis also designs the classifier structure of a global classifier combined with a local classifier and introduces the spatial self-attention mechanism and pixel self-attention mechanism to extract richer features of bone marrow microscopy images. Through experimental validation, this thesis's method outperforms other semi-supervised classification methods in terms of classification accuracy with limited labeling and achieves the classification of four types of hematologic malignancies.

Key Words: Bone marrow micrographs, Image classification, Stain normalization, Self-attention mechanism, Semi-supervised

Semi-Supervised Stain Normalization Network Based on Self-Attention Mechanism

Shao Hong[1], Zhou Yang[1], Cui Wencheng[1]

1. *School of Information Science and Engineering, Shenyang Institute of Technology, Shenyang 110870, P.R. China*

【Abstract】 Histocytopathology is the gold standard for disease diagnosis. However, different laboratory preparation methods and differences in scanning equipment can lead to changes in the color of pathology images. The change in color can introduce undesirable errors when the images are diagnosed by pathologists or when these pathology images are used to train deep learning models In this paper, a semi-supervised staining normalization network based on a self-attention mechanism is introduced. The network incorporates a self-attention mechanism for generating images with more detailed features while maintaining structural consistency of pathology features during the stain normalization process. In addition, the network combines supervised learning of the target image and unsupervised learning of the source image using consistency regularization loss to keep the images generated by the dual decoders consistent. Experimental results show that the network exhibits good performance in the pathology image staining normalization task. In the task of classification subsequent to image staining normalization, the performance is improved by 4% to 10% compared to the benchmark algorithm that does not use the normalization method.

【Key words】 Histiocytopathology images; Self-attention Mechanism; Semi-Supervised; Staining Normalization; Generative Adversarial Network

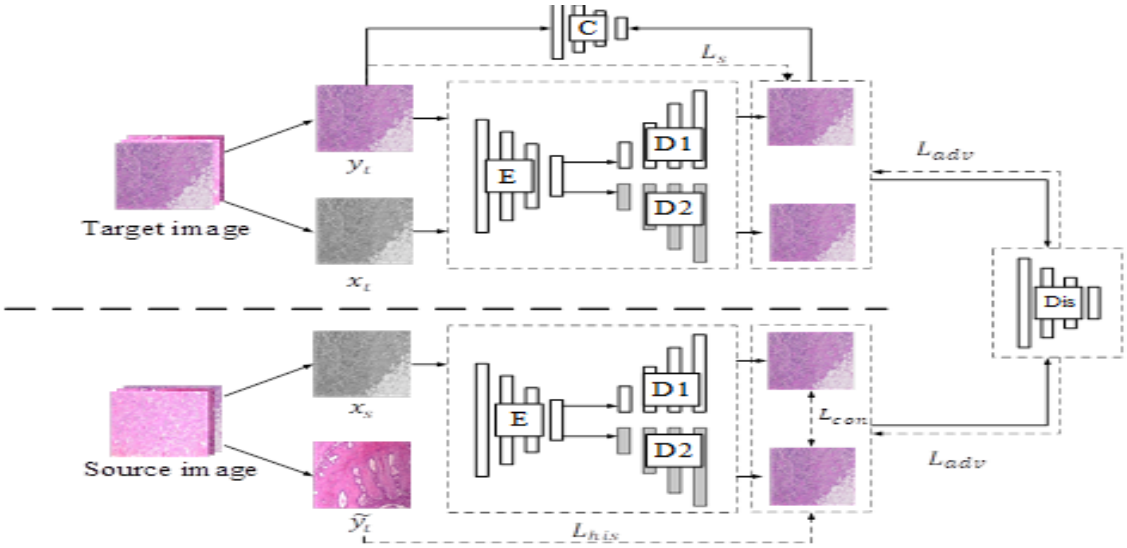
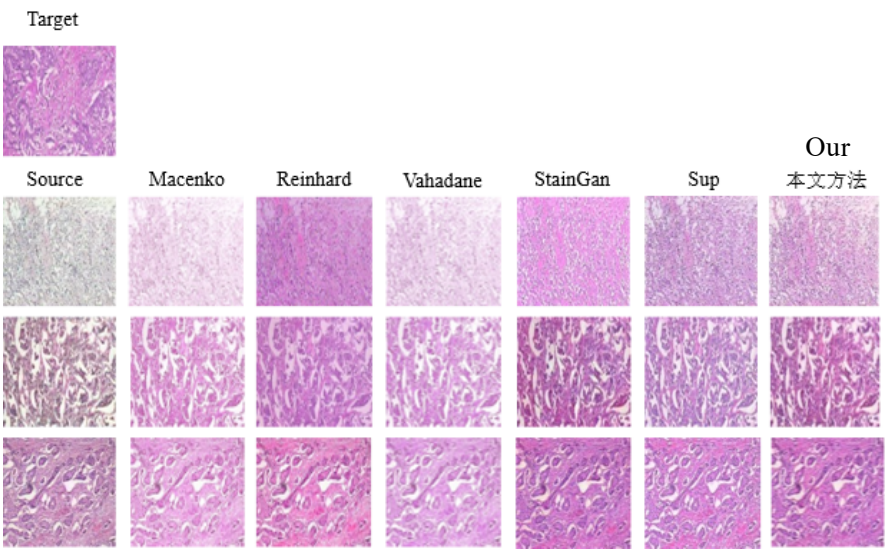


Fig. 3.2 Stained normalized model structure



通信作者：周阳，Email：zhouyangs@foxmail.com

Yang Zhou*, Hong Shao. PLE-GAN: Pseudo-label enhanced generative adversarial networks for medical image classification[C]. 2024 3rd International Symposium on Computer Applications and Information Systems(ISCAIS 2024). Xiamen:SPIE,2023, 224031123531928922.

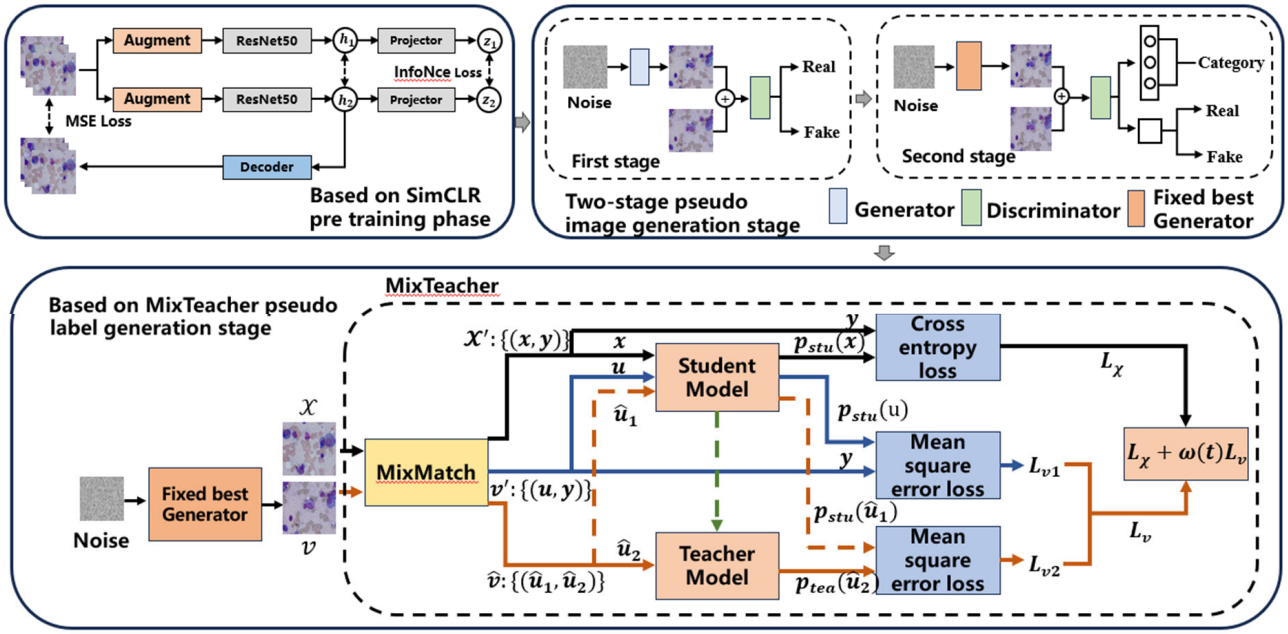
PLE-GAN: Pseudo-label enhanced generative adversarial networks for medical image classification

Yang Zhou*, Hong Shao
School of Computer Science and Technology, Shenyang University of Technology, Shenyang, 110000, China
*Corresponding author: zhouyangs@foxmail.com

ABSTRACT

Currently, semi-supervised learning methods have become popular in medical image processing tasks, this paper proposes a medical image classification method based on a semi-supervised pseudo-label augmented generative adversarial network. This method combines the generative adversarial network and pseudo-label generation technology, introduces the FID evaluation index to train the generator with good performance for the problem of poor classification performance of GAN, and uses only a small number of real images to generate a sufficiently large number of pseudo-images for model training, which is combined with the pseudo-label generation method of teacher-student network introduced based on MixMatch to categorize the medical images. Through experimental validation, this paper's method outperforms other semi-supervised classification methods in terms of classification accuracy with limited labeling and achieves classification on three publicly available medical image datasets. In addition, we conducted ablation experiments to verify the algorithm's effectiveness.

Keywords: Generative adversarial networks, Semi-supervised classification, Pseudo-labeling, Medical image



Third International Symposium on Computer Applications and Information Systems (ISCAIS 2024), edited by Hongzhi Wang, Wenlong Li, Proc. of SPIE Vol. 13210, 132100H
© 2024 SPIE · 0277-786X · doi: 10.1117/12.3034922

National Copyright Administration of the People's Republic of China

Computer Software Copyright Registration Certificate

Certificate No.: RZD Zi No. 11402509

Software Title: Bone Marrow Aspiration Smear Staining Normalization Software

[Abbreviation: Staining Software]

V1.0

Copyright owner: Zhou Yang; Shao Hong; Shenyang University of Technology

Development Completion Date: May 6, 2023

Date of first release: Unpublished

Mode of Acquisition of Rights: Original acquisition

Scope of Rights: Full rights

Registration No.: 2023SR0815338

Upon the verification by China Copyright Protection Center, the above matters are registered in accordance with the provisions of the Regulations on the Protection of Computer Software and the Measures for the Registration of Computer Software Copyright.

Special Seal for Registration of Computer Software Copyright of the National Copyright Administration of the People's Republic of China (sealed)

No.13202037 July 6, 2023

I, Ji Xiangdi, translator of E-TRANSTAR (Beijing) Information Technology Co., Ltd., confirm this is a true and accurate translation of the original document.
CATTI Certificate No.: 201906009370001040 Organization: E-TRANSTAR (Beijing) Information Technology Co., Ltd.
Tel: 010-60537024 Email: 2355529118@qq.com
Organization Address: 201-212057, Zone 6, Pinggu Park, Zhongguancun Science and Technology Park, Pinggu District, Beijing
Signature: Date of Translation: July 15, 2025

Ji Xiang di

E-Transtar (Beijing) Information Technology Co., Ltd. 翻译专用章 Valid for Translation Only

National Copyright Administration of the People's Republic of China
Computer Software Copyright Registration Certificate (original document)

中华人民共和国国家版权局

计算机软件著作权登记证书

证书号： 软著登字第11402509号

软 件 名 称： 骨髓穿刺涂片染色归一化软件
[简称：染色软件]
V1.0

著 作 权 人： 周阳;邵虹;沈阳工业大学

开发完成日期： 2023年05月06日

首次发表日期： 未发表

权利取得方式： 原始取得

权 利 范 围： 全部权利

登 记 号： 2023SR0815338

根据《计算机软件保护条例》和《计算机软件著作权登记办法》的规定，经中国版权保护中心审核，对以上事项予以登记。

中华人民共和国国家版权局
计算机软件著作权
登记专用章

No. 13202037

2023年07月06日

China National Intellectual Property Administration

100080 Beijing Fangyuan Jiahe Intellectual Property Agency Co., Ltd. 11th Floor, Room 1104-2, No. 8, Caihefang Road, Haidian District, Beijing Wang Yuesong (010-61943059)	Issuing Date: August 9, 2024

Application No. or Patent No.: 202410612463.7 Document No.: 2024080900448160

Applicant or Patentee: Shenyang University of Technology

Invention Title: Staining - Normalization Image Processing Method, Device and Product Based on Improved GAN

Notice about the Invention Patent Application for Substantive Examination Stage

As for the above patent application, according to the substantive examination request proposed by the applicant, the patent application conforms to the regulations in Article 35 of Patent Law and Article 113 of the Rules for the Implementation. The patent application has entered the substantive examination stage.

Note:

1. According to the regulations in Item (I) of Article 57 of the Rules for the Implementation of the Patent Law of the People's Republic of China, the invention patent applicant may propose to modify the application of invention patent within 3 months after receiving the notice.

2. Requirements for modifying the format of application documents:

Where the claim is amended, the corresponding replacement entry of the claim shall be submitted, and if that refers to a reference relationship of the claim, the corresponding claim shall be replaced together. If the applicant needs to delete some of the claims, the applicant shall submit partial arranged claims in consecutive number.

The corresponding specification replacement paragraph shall be submitted for the amendment of the specification, and the paragraph number shall not be added and deleted, and only in the contents to be amended, a whole paragraph can be replaced. The proposed contents can only be added in a certain paragraph. The paragraph number shall be remained when the one whole paragraph is required to be deleted, and the following content shall be indicated behind the paragraph number: The words "this paragraph is deleted". The paragraph number shall prevail in accordance with the paragraph number of the specification which is passed back or under publication/authorized publication of China National Intellectual Property Administration.

The corresponding replacement figure shall be submitted with respect to the amendment of the figure attached to the specification, taking the figure as a unit.

If the text of the abstract needs to be modified, the corresponding replacement page shall be submitted. If the figure attached to the abstract is modified, it shall be re-designated.

At the same time, the applicant shall specify the claims, paragraph number, figures and pages involved in the modification in the certificate of correction or statement of opinions.

Reviewer: Automatic Review

Contact No.: 010-62356655

Examination Department: Preliminary Examination and Process Management Department

Business Seal for Patent Examination of China National Intellectual Property Administration (sealed)

210307 If it is paper document application, please mail letter in reply to: Patent Office of the China National Intellectual Property Administration, No.6 Xitucheng Road, Jimenqiao, Haidian District, Beijing Postal Code: 100088

2023.03 For electronic application, please submit relevant documents in electronic form through the Patent Business Handling System. Unless otherwise specified, documents submitted in other forms, such as paper, shall be regarded as not submitted.

I, Ji Xiangdi, translator of E-TRANSTAR (Beijing) Information Technology Co., Ltd., confirm this is a true and accurate translation of the original document.
CATTI Certificate No.: 201906009370001040 Organization: E-TRANSTAR (Beijing) Information Technology Co., Ltd.
Tel: 010-60537024 Email: 2355529118@qq.com
Organization Address: 201-212057, Zone 6, Pinggu Park, Zhongguancun Science and Technology Park, Pinggu District, Beijing
Signature: Date of Translation: July 15, 2025

Ji Xiang di

翻译专用章 Valid for Translation Only